Types of gear shift mechanisms

1. Floor gear shifting Mechanism
2. Steering Gear Shifting Mechanism

Floor gear shifting mechanism

If the gear shifting mechanism is located at the top of the gearbox, then it is called as the floor gear shifting mechanism

Steering gear shifting mechanism

If the gear shifting mechanism is located at the sideway of the steering column, then it is called as the steering gear shifting mechanism

3.10 Gear box lubrication

1. In the gear box, lubricating oil of SAE 90 has to be filled.
2. The lubricating oil must be filled until the lay shaft is (dipped) covered by the oil. The lubricating oil must be highly viscous.
3. When the gears are rotating, the lubricating oils is sprayed on all the parts of the gear box and lubricated.
4. After 1000 kilo meters of the vehicles run lubricating oil top up must be done.
5. After 10000 kilometers of the vehicle the lubricating oil must be completely replaced by the new lubricant.
6. To pour the oil filler cap is provide on the top of the gear box. An air vent is also provide to allow the atmospheric air to enter.
7. A drain plug is provided at the bottom of the gear box to drain the lubricating oil.

3.11 Transfer case

The arrangement used for transmitting the engine's power to all the four wheels is called as the transfer case.

The transfer case is fitted on the vehicles with four wheel drive. This arrangement is generally used in military vehicles and jeeps. A photographic view of an automobile transfer case can be seen in Figure 3.11.1.



Figure 3.11.1 Photographic vie of an automobile transfer case

Construction

In the main shaft of the gear box two gears are present. Between the two gears a sliding gear is present. The gear in the main shaft is in mesh with the idler gear. With this front axle drive gear and rear axle drive gear are connected. A clutch is fitted with the front axle drive gear. The important parts of the transfer case can be seen in Figure 3.11.2.

Operation

With the help of the gear shift mechanism when the sliding gear in the main shaft is shifted to right or left, through the idler gear the power is transmitted to the front or rear wheels with low or high speeds. In addition to this, when there is no need for drive power to the front wheels, with the help of the clutch arrangement the drive power can be disengaged. If necessary the four wheel drive can be also obtained.

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